Practice Problem

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Gases

e-book lesson: 12-1 to 12-6

Objectives: After this lesson, you should be able to…

- Describe properties of gases and compare to liquids and solids.
- Describe the key postulates of KMT and explain how KMT relates to behavior of gas particles.
- Connect the effect of temperature on the average kinetic energies and the root-mean-square speed of gases.
- Interpret a Boltzmann distribution of molecular speeds and match Boltzmann curves to gas samples of various temperatures and molar masses.
- Calculate the root mean square speed of a gas at any given temperature.
- Define effusion and diffusion, and compare effusion or diffusion rates of two gases.

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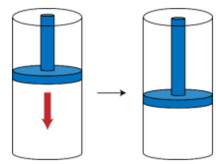
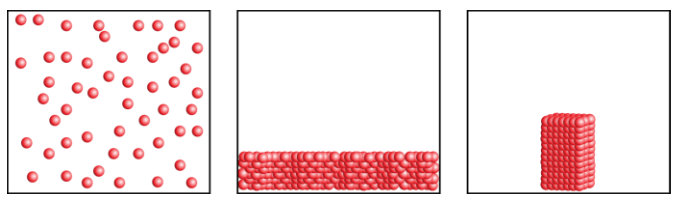
From the PLA: Gases have unique properties

Gases are a collection of atoms or molecules that…

- assume the shape and size of their container (volume = V)
- exert a force called pressure (P)
- have low densities
- are compressible





gas liquid solid

On the molecular level, what gives rise to these properties?

From the PLA: The kinetic molecular theory of gases: molecules are in constant motion

Five key postulates of kinetic molecular theory:

1. Molecules move in straight lines; their direction is random.
2. Molecules are small. The volume of the molecules is much smaller than the volume of the container.
3. Molecules do not attract or repel each other (no IM forces).
4. Molecules experience elastic collisions.
5. The <u>mean</u> (average) kinetic energy (ε) is proportional to temperature (in Kelvin)

$$\varepsilon \propto T$$

Kinetic molecular theory demonstration

https://genchem.science.psu.edu/apps/kinetic_theory

1. What do you notice about the speeds of the helium atoms? Are they all the same?

2. If we increase the temperature, what happens to the speeds?

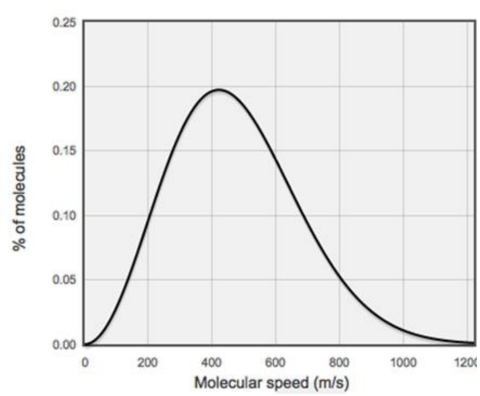
3. Let's add in some neon. How do the speeds of the neon atoms compare to the speeds of the helium atoms?

4. Let's reduce the size of the container. What happened to the speed of the atoms? What happened to the pressure?

Temperature is related to the speeds of molecules

$$T \propto \varepsilon = \tfrac{1}{2} m \cdot u^2$$

ε = average kinetic energy
m = mass of molecule (in kg)
u = root mean square (rms) speed of molecule



Molecules are moving at different speeds.

Within a large collection of molecules, the energies (and speeds) will follow a well-known pattern called the Maxwell-Boltzmann distribution.

Root mean square speed is related to both molar mass of the gas and temperature

$$u = \sqrt{\frac{3RT}{M}}$$

u = rms speed
M = molar mass (in kg/mol)
T = temperature (Kelvin)
R = 8.314 $J \cdot mol^{-1} \cdot K^{-1}$

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Distribution of molecular speeds at different temperatures

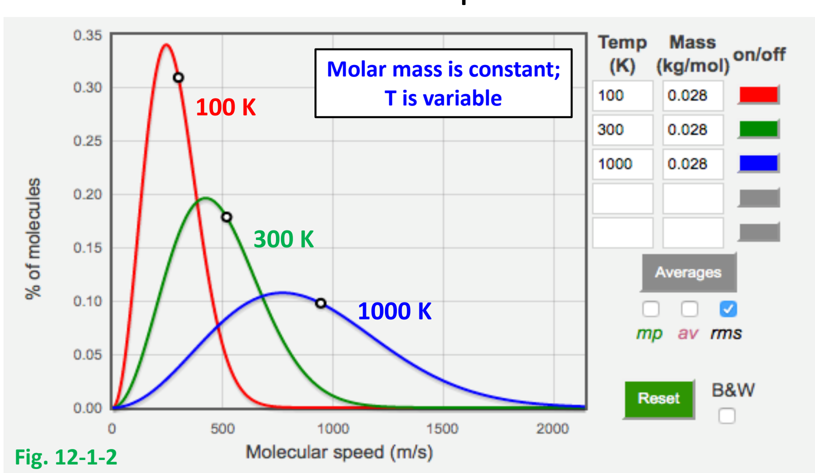

Fig. 12-1-2

As T increases: - Fewer molecules move slowly (more molecules move quickly).

- The rms speed increases (black dot on curve = rms speed).

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The distributions below are for different molecules at the same temperature

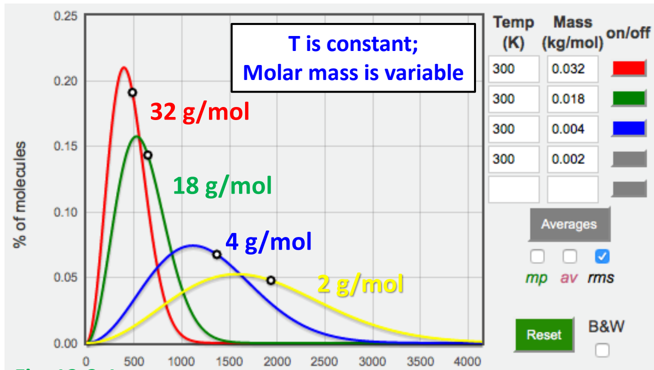

Fig. 12-2-1

At the same T, different gases have the same average kinetic energy, but different rms speeds.

On average: lighter gases move faster and heavier gases move slower.

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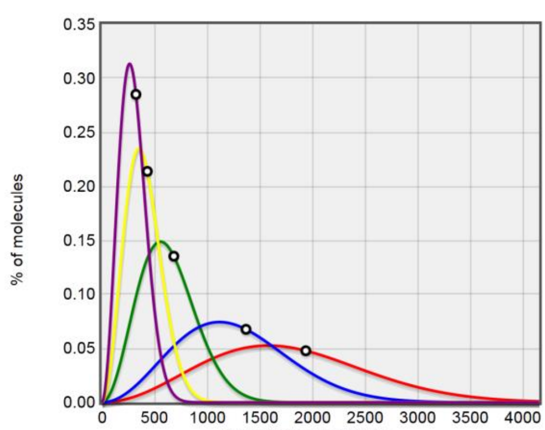

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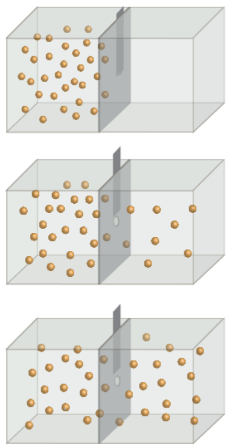
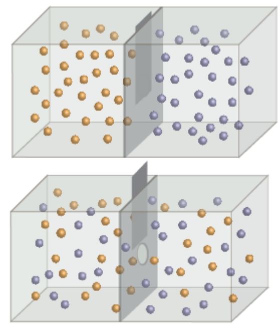
Macroscopic motion of gases





Diffusion: even distribution of all gases throughout a volume of space.

Effusion: leakage of gas through a small opening.

Rates of effusion and diffusion

The rate of effusion (or diffusion) is proportional to rms speed. If we measure effusion rates for two gases under the same conditions we can express this as a ratio:

$$\frac{r_1}{r_2} = \frac{u_1}{u_2} = \sqrt{\frac{\left(\frac{3RT}{M_1}\right)}{\left(\frac{3RT}{M_2}\right)}} = \sqrt{\frac{M_2}{M_1}}$$

We can use Graham's law to gather information about the molecular speed or molar mass of a gas.

Graham's Law $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

Light molecules effuse/diffuse faster than heavy molecules.

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Pressure is caused by molecular collisions

Pressure is the force caused by molecules striking a surface per unit area.

Pressure is defined as force per unit area:

Force is measured in Newtons (N), area is measured in meters squared (m^2)

SI unit is the Pascal (Pa): $1\ \frac{N}{m^2} = 1\ Pa$

Standard atmospheric pressure:

1 atm = 1.01325×10^5 Pa = 1.01325 bar

1 atm = 14.7 lb/in^2

1 atm = 760 torr = 760 mm Hg

(actual atmospheric pressure
varies with altitude and weather)